Lesson plan · for the teacher

Matching teachers and classes

Optimization · Level 3 · Lesson 4

Time	Materials
40 min	Board and chalk; A score table on the board: pupils by jobs; Small cards for the jobs

Learning goals

- Write an assignment as yes-or-no choices, one per row and column.
- Find the best total, and the plan with the best worst match.
- Ask who wrote the scores, and decide the rule together.

Activity: Fair class jobs

1. Warm-up (5 min): Ask how class jobs are given out now and whether pupils think it is fair. Write their ideas of “fair” on the board.
2. Score (8 min): Write five jobs across the top of a table and five volunteers down the side. Each volunteer scores every job from 1 (hate it) to 5 (love it).
3. In turn (7 min): Volunteers choose one job each, in order of the register. Add up the total score and circle the lowest score.
4. Better plans (15 min): In groups, pupils look for the plan with the highest total, then for the plan whose lowest score is as high as possible. Each group writes its two plans with their total and lowest score.
5. Wrap-up (5 min): Compare the plans and vote on the rule for this term. Explain it to the volunteers before announcing the jobs.

Keep it to five volunteers and five jobs so the table fits the board. Scores of 1 to 5 keep the sums easy.

Discussion

- Why did choosing in turn give a lower total?
- Which plan was best for the person with the lowest score?
- Who should write the scores, and who should see them?

Exit ticket

- Write one rule for sharing class jobs fairly, and what it might cost.

Try it at school tomorrow

Share out five class jobs (board, plants, books, water, bell). Each pupil scores every job from 1 to 5. First let them choose in turn, then look together for the plan with the best total, and for the plan where nobody gets a job they scored 1. Which would the class choose, and why?

Practice questions · for you

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1 **The best-total plan gives Zainab, a new teacher, the class nobody wanted. Her scores were written by the deputy head, and nobody asked her. What is the fair next step?**

◯ Ask her, check her scores, and solve again with the agreed rule ◯ Keep it: the solver maximised the total

◯ Swap her with the most senior teacher

2 **Letting teachers choose classes in turn always gives the best total score.**

◯ True ◯ False

3 **What does this program print?**

```
score = [[9, 8, 6], [9, 2, 1], [6, 9, 5]]
pairs = [0, 1, 2]  # teacher t takes class pairs[t]
print(sum(score[t][pairs[t]] for t in range(3)))
```

◯ 16 ◯ 24 ◯ 6

4 **Three teachers, three classes, and a score for each pair. The model already says each teacher takes one class and each class gets one teacher. Write the goal: the biggest total score of the chosen pairs. The program prints the total.**

```
from ortools.sat.python import cp_model
score = [[9, 8, 6], [9, 2, 1], [6, 9, 5]]
n = range(3)
m = cp_model.CpModel()
x = [[m.new_bool_var("") for c in n] for t in n]
for i in n:
    m.add_exactly_one(x[i])
    m.add_exactly_one([x[t][i] for t in n])
# the goal: the biggest total score
solver = cp_model.CpSolver()
solver.solve(m)
print(int(solver.objective_value))
```

Write your program:

5 **The model already gives each class exactly one teacher. Complete it so that the worst match is as good as possible: each teacher takes exactly one class, and every teacher's chosen score is at least w.**

```
for t in range(3):
    m. _____(x[t])
w = m.new_int_var(1, 9, "worst")
for t in range(3):
    m.add(sum(score[t][c] * x[t][c]
              for c in range(3)) _____ w)
m. _____(w)
```

Word bank: `>=` `add_all_different` `minimize` `<=` `add_exactly_one` `maximize`

✎ **How are classes, duties or rooms shared out at your school now? Who would score better, and who worse, under a best-total rule?**

Check your answers

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1 **Ask her, check her scores, and solve again with the agreed rule**

A model is only as fair as its data. Zainab's scores are someone else's opinion of her, so check them with her, agree the rule (the best total, or a floor for the worst match), and solve again.

2 **False**

An early choice can take someone else's only good option: in the lab, choosing in turn gives 29 against the best total of 39, because Amina takes the maths class Karim needed.

3 **16**

Teacher 0 takes class 0 (9), teacher 1 class 1 (2) and teacher 2 class 2 (5): 9 + 2 + 5 = 16.

4

```
from ortools.sat.python import cp_model
score = [[9, 8, 6], [9, 2, 1], [6, 9, 5]]
n = range(3)
m = cp_model.CpModel()
x = [[m.new_bool_var("") for c in n] for t in n]
for i in n:
    m.add_exactly_one(x[i])
    m.add_exactly_one([x[t][i] for t in n])
m.maximize(sum(score[t][c] * x[t][c] for t in n for c in n))
solver = cp_model.CpSolver()
solver.solve(m)
print(int(solver.objective_value))
```

The best total is 24: teacher 1 takes class 0 (9), teacher 2 class 1 (9) and teacher 0 class 2 (6).

5

```
for t in range(3):
    m.add_exactly_one(x[t])
w = m.new_int_var(1, 9, "worst")
for t in range(3):
    m.add(sum(score[t][c] * x[t][c]
              for c in range(3)) >= w)
m.maximize(w)
```

Exactly one class per teacher, each chosen score at least w (expression on the left, then >=), and w as large as possible: the worst match rises as far as the table allows.